

Endpoint Metrics Bridge for Anonymous Overlays: Odds Inflation, Pointwise Maximal Leakage, and Maximal Leakage

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Abstract

Several papers in this archive budget anonymity using different endpoint metrics: posterior-odds inflation (Anonymity A), pointwise maximal leakage (PML), and maximal leakage (MaxL) in the “Boss Fight” repeated-use regime. This short synthesis note pins down the exact algebraic relationships among these endpoints (under a fixed prior), and records the two sizing rules most often needed by operators and referees: (i) MaxL is the log moment generating function of realized odds inflation; and (ii) keeping exact-guess probability below α for a K -way secret requires total MaxL at most $\log(\alpha K)$ (nats) or $\log_2(\alpha K)$ (bits). The goal is non-redundancy: other papers may cite this note instead of re-deriving these bridges.

Scope. Pin down the algebraic relationships among endpoint anonymity metrics used across the archive (odds inflation, PML, MaxL) and provide sizing rules for repeated use.

Imports. We import the odds-inflation definition from Anonymity A and the repeated-use “Boss Fight” framing.[4, 5] We also treat the backbone optimal distinguishing bounds as imported (see [[2026.01.23 - Mathematics: Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays]]).

Exports. Equivalences and inequalities relating odds inflation, PML, and MaxL (under a fixed prior), plus the standard $\log(\alpha K)$ identification sizing rule used throughout the series.

Artifact policy. Source-only; supporting artifacts are available on request. See [1].

1 Setup (one secret, one observation, one prior)

Let $S \in \mathcal{S}$ be a discrete secret with prior π , and let $Y \in \mathcal{Y}$ be an observation. Write P_s for the law of Y given $S = s$ and write the public (mixture) marginal

$$P(y) \stackrel{\text{def}}{=} \sum_{s \in \mathcal{S}} \pi(s) P_s(y).$$

All quantities below are properties of the channel $\{P_s\}_{s \in \mathcal{S}}$; only the odds-inflation / pointwise forms reference the prior through P .

2 Realized odds inflation equals pointwise maximal leakage

Definition 1 (Realized odds inflation / pointwise Bayes factor). *For any y with $P(y) > 0$, define*

$$\ell(y) \stackrel{\text{def}}{=} \max_{s \in \mathcal{S}} \log \frac{\pi(s | y)}{\pi(s)} = \max_{s \in \mathcal{S}} \log \frac{P_s(y)}{P(y)}.$$

Remark 1 (Name alignment). *Many references define pointwise maximal leakage (PML) at outcome y as $\log \max_s \frac{P_s(y)}{P(y)}$ (choice of log base varies). Thus, under a fixed prior, PML at y is exactly $\ell(y)$. See [3] for operational interpretations.*

3 MaxL is the log-moment of realized odds inflation

Definition 2 (Maximal leakage (nats) [2]). *For discrete S, Y , define*

$$\text{MaxL}(S \rightarrow Y) \stackrel{\text{def}}{=} \log \sum_{y \in \mathcal{Y}} \max_{s \in \mathcal{S}} P_s(y).$$

(Using $\log_2$ gives MaxL in bits.)

Lemma 1 (Moment identity). *Let $Y \sim P$ be drawn from the mixture marginal. Then*

$$\sum_{y \in \mathcal{Y}} \max_s P_s(y) = \mathbb{E}_{Y \sim P}[\exp(\ell(Y))].$$

Equivalently, in nats,

$$\text{MaxL}(S \rightarrow Y) = \log \mathbb{E}_{Y \sim P}[\exp(\ell(Y))].$$

Proof. By definition, $\mathbb{E}_P[\exp(\ell(Y))] = \sum_y P(y) \max_s \frac{P_s(y)}{P(y)} = \sum_y \max_s P_s(y)$. Taking log gives the identity. $\square$

Proposition 1 (Two immediate corollaries). *1. (Worst-case dominates.) $\text{MaxL}(S \rightarrow Y) \leq \sup_y \ell(y)$ (log of an average is at most the max).*

2. (Typical vs worst-case.) If $\ell(Y) \leq \varepsilon$ almost surely under P , then $\text{MaxL}(S \rightarrow Y) \leq \varepsilon$. More generally, tail envelopes on $\ell(Y)$ translate into bounds on MaxL via the moment identity.

4 Sizing rules for exact identification

The Boss-Fight series uses MaxL as a repeated-use identification budget. The following bound is the standard “back-of-the-envelope” sizing rule.

Lemma 2 (Exact-guess probability bound). *For any prior π ,*

$$P_{\text{guess}}(S | Y) \leq \exp(\text{MaxL}(S \rightarrow Y)) \cdot \max_s \pi(s),$$

where $P_{\text{guess}}(S | Y) \stackrel{\text{def}}{=} \mathbb{E}[\max_s \pi(s | Y)]$. In particular, if S is uniform on K secrets, then $P_{\text{guess}}(S | Y) \leq \exp(\text{MaxL})/K$.

Proof. For any outcome y , $\max_s \pi(s | y) = \max_s \pi(s) \frac{P_s(y)}{P(y)} \leq (\max_s \pi(s)) \exp(\ell(y))$. Taking expectation under P and using the moment identity yields the claim. $\square$

Remark 2 (Window budgeting). *If an operator wants $P_{\text{guess}} \leq \alpha$ for a uniform K -way secret, it suffices to enforce*

$$\text{MaxL}(S \rightarrow Y_{1:T}) \leq \log(\alpha K) \quad (\text{nats}) \quad \text{or} \quad \text{MaxL}(S \rightarrow Y_{1:T}) \leq \log_2(\alpha K) \quad (\text{bits}).$$

This is the sizing rule used in the Boss-Fight dial sheet.

How this note interacts with the rest of the archive

- Odds-inflation anonymity (Anonymity A) develops composition rules and converts common contracts (TV equalization, moment bounds) into bounds on $\ell(y)$ (and hence on MaxL via this note) [4].
- The Boss-Fight series uses MaxL/PML as the endpoint in repeated-use identification regimes and compiles MaxL targets into system knobs [5, 6].
- The trace-interface and threat/window notes standardize what the observation is and over what repetition window the above budgets apply [7, 8].

References

- [1] Anonymous. Artifact and Disclosure Policy for the One-True-Archive (Source-Only Public Release). Series synthesis note (Synthesis 6), 2026. (This archive.)
- [2] I. Issa, A. B. Wagner, and S. Kamath. An operational approach to information leakage. *IEEE Trans. Inf. Theory*, 66(3):1625–1657, 2020.
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- [4] Anonymous. Odds-Inflation Anonymity: Posterior Odds Budgets, Composition, and Operational Limits. Anonymity series Paper A, 2026. (This archive.)
- [5] Anonymous. Boss Fight Budgets: Maximal Leakage as a Repeated-Use Anonymity Contract. Boss Fight series Paper A, 2026. (This archive.)
- [6] Anonymous. AnonDHT in the Boss Fight regime: compiling a MaxL budget into lookup knobs. Boss Fight series Paper B, 2026. (This archive.)
- [7] Anonymous. Trace Interfaces for Anonymous DHTs: Observation Tiers, Committee-Contact Traces, and Exposure Normal Forms. Series synthesis note, 2026. (This archive.)
- [8] Anonymous. Threat Models and Repetition Windows for Anonymous DHTs: a short declaration checklist. Series synthesis note, 2026. (This archive.)
- [9] Anonymous. [[2026.01.23 - Mathematics: Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays]]. Published mathematics note, 2026. (This archive.)